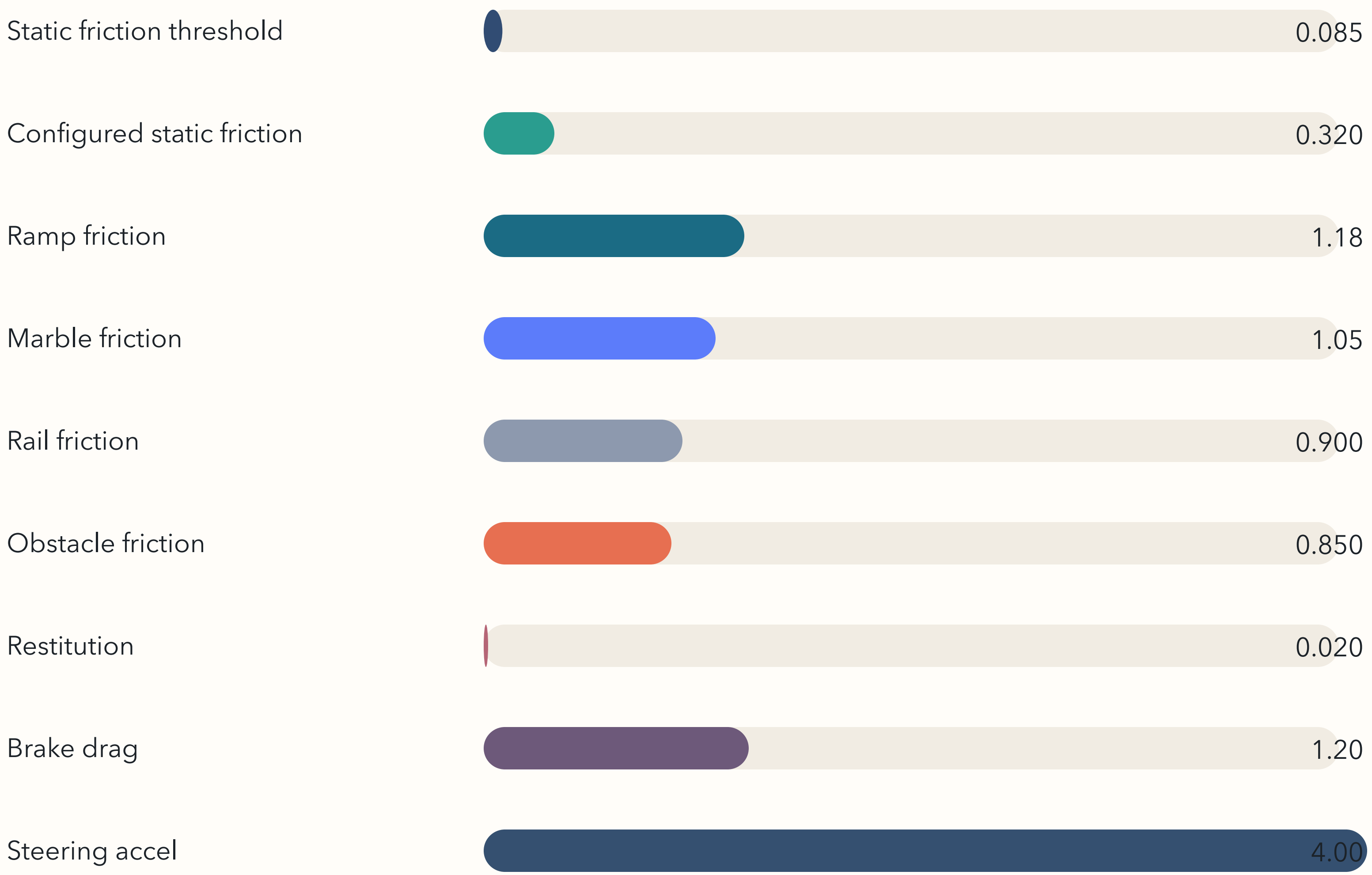
Figure 3. Physics tuning summary

The project combines physically motivated limits with deliberately game-tuned response values.

SimulationConfig comparison

Bars share a common scale so the deliberately tiny restitution remains visually obvious.



Pure-rolling check

The static friction threshold is computed from the incline angle.

$$\mu_{s,req} = (2/7)\tan(\theta)$$

$\theta = 16.52°$

$\mu_{s,req} = 0.085$

configured $\mu_s = 0.320$

Rolling supported

What matters in play

These choices are consistent with the report's simulation-versus-arcade framing.

- Low restitution keeps impacts from feeling pinball-like.
- Brake drag is exponential rather than a hard velocity clamp.
- Steering is a lateral force, not a yaw rotation hack.
- Ramp and marble friction exceed the pure-rolling threshold.